SUMIT KUMAR PATEL

AI/ML Engineer | Backend Engineer

+91-7008404037 | sumitpatelofficial1@gmail.com | linkedin.com/in/sumit-patel-8845511ab/ | github.com/Sumitpatel29

PROFESSIONAL SUMMARY

AI/ML and Backend Engineer with hands-on experience building LLM-powered conversational AI agents, real-time voice systems, telephony integrations, and production backend services. Experience integrating large language models (OpenAI GPT-4o-mini, Sarvam AI) into multi-turn conversational workflows, including conversation-flow design, state management, and automated lead classification. Solid foundation in classical Machine Learning (TF-IDF, Logistic Regression, SVM, Random Forest) and on-device deep learning with TensorFlow Lite. Proficient in Python, Java, FastAPI, MongoDB, SQL, Redis, and Docker, with strong fundamentals in Data Structures and Algorithms, OOP, Design Patterns, and SDLC.

TECHNICAL SKILLS

- AI / ML / LLM:Claude Code, OpenAI GPT-4o-mini, Sarvam AI, Conversational AI Agents, NLP, Machine Learning (TF-IDF, Logistic Regression, SVM, Random Forest), TensorFlow Lite, Multimodal Data Fusion
- Programming Languages: Python, Java, JavaScript
- Backend: FastAPI, Spring Boot, Spring MVC, JDBC, J2EE, JSP, Servlets, Hibernate, JPA
- Databases (SQL & NoSQL): MongoDB, MySQL, PostgreSQL
- Caching and Queues: Redis, arq
- Telephony and Communication: Plivo, SIP, WebRTC, Meta WhatsApp Cloud API, LiveKit
- APIs and Security: RESTful APIs, Webhooks, JWT, OAuth2, HMAC, API Authentication
- DevOps and Deployment: Docker, nginx, systemd, Linux
- Testing: pytest, Unit Testing, Mocking
- Tools: Git, GitHub, VS Code, Eclipse, NetBeans, Postman
- Software Engineering: DSA, OOP, OOAD, Design Patterns, SDLC, Agile

PROFESSIONAL EXPERIENCE

AI/Backend Engineer — Upskalor Technologies LLP *(2026 – Sep 2026)*

- Developed backend services and AI-powered communication systems using Python, FastAPI, MongoDB, Redis, Docker, LiveKit, Plivo, Meta WhatsApp Cloud API, OpenAI, and Sarvam AI.
- Built and deployed real-time voice and communication workflows involving telephony, webhooks, asynchronous processing, authentication, and third-party API integrations.

Sahayak Odia Voice Agent

Technologies: Python, LiveKit, WebRTC, SIP, Plivo, Sarvam AI, OpenAI, Docker, Redis, nginx

- Built a self-hosted, real-time conversational AI agent capable of understanding and responding in Odia.
- Integrated LiveKit for real-time media processing and WebRTC/SIP-based communication for inbound and outbound calls through Plivo.
- Integrated Sarvam AI speech-to-text and text-to-speech services with GPT-4o-mini as the conversational reasoning layer of the agent.
- Deployed the voice agent on a VPS using Docker Compose, nginx, Redis, and systemd.
- Configured firewall rules and SIP/RTP networking required for production telephony communication.
- Tuned audio, latency, and voice-processing parameters to improve real-time conversational performance.
- Documented production troubleshooting procedures covering common audio, latency, SIP, and deployment issues.

Missed-Call to WhatsApp Follow-up Platform

Technologies: Python, FastAPI, Redis, arq, Plivo, Meta WhatsApp Cloud API, Fernet, pytest

- Developed a multi-tenant FastAPI platform that detects missed calls through Plivo webhooks and automatically sends WhatsApp template messages through the Meta Cloud API.
- Implemented asynchronous background processing using arq and Redis to separate webhook handling from message delivery.
- Implemented idempotency and deduplication mechanisms to prevent duplicate WhatsApp messages when webhook events are retried.
- Secured tenant-specific Meta API credentials using Fernet encryption.

- Implemented Plivo webhook signature verification using HMAC to validate incoming requests.
- Implemented timezone-aware business-hour processing and opt-out handling for communication compliance requirements.
- Developed 300+ automated pytest tests using mocks and isolated test components without requiring external services.
- Implemented a staged rollout process with a send-allowlist safety gate to control message delivery during deployment and testing.
- Built funnel and failure-tracking APIs for monitoring message delivery and communication outcomes.

WhatsApp Lead Qualification Bot (Conversational AI Agent)

Technologies: Python, FastAPI, MongoDB, Beanie ODM, Meta WhatsApp Cloud API, JWT, OAuth2

- Developed a conversational WhatsApp lead qualification agent with a structured 10-question flow in Odia.
- Implemented a conversational state machine to maintain agent progress and context across multiple questions and responses.
- Implemented lead classification into HOT, WARM, and COLD categories based on qualification responses.
- Integrated Google Sheets export for transferring qualified lead information to business workflows.
- Implemented Meta webhook signature verification using X-Hub-Signature-256.
- Implemented JWT-based authentication and OAuth2-based authentication flows for the management dashboard.
- Used MongoDB with Beanie ODM for persistence and management of lead and conversation data.
- Developed automated tests covering API endpoints, business logic, authentication, and conversational workflows.

Outbound Calling and Campaign Services

Technologies: Python, FastAPI, APScheduler, MongoDB, Streamlit, Plivo

- Developed FastAPI services for immediate and scheduled outbound calling workflows.
- Implemented scheduled call execution using APScheduler and maintained task execution metrics in MongoDB.
- Built a Streamlit-based control dashboard for managing and monitoring calling workflows.
- Developed an auto-dialer campaign API with lead tracking and CSV import/export functionality.
- Implemented backend APIs for managing campaign leads, scheduled calls, and call-related task information.

PROJECTS

Mental Health Analysis *12/2024 – 01/2026*

Technologies: Android Studio, Python, TensorFlow Lite | NLP, Machine Learning, Mobile Sensing

- Designed and developed a privacy-preserving mental health monitoring system integrating NLP-based text analysis with smartphone behavioral indicators such as activity, sleep, and mobility patterns.
- Implemented lightweight machine learning models including TF-IDF, Logistic Regression, SVM, and Random Forest for on-device mental health screening.
- Applied multimodal data fusion techniques to combine linguistic and behavioral signals while supporting local processing and privacy-focused deployment.

Banking System *12/2024 – 01/2025*

Technologies: Eclipse IDE, MySQL Workbench | Core Java, Hibernate, MySQL, Tomcat, JDBC

- Designed and developed a Java-based banking application implementing account creation, user authentication, balance inquiry, and money transfer operations.
- Implemented database connectivity using JDBC and Hibernate with transaction management and SQL-based data operations.
- Added input validation and exception handling to improve application reliability and data integrity.

EDUCATION

Master of Computer Applications — Institute of Technical Education and Research, Bhubaneswar, Odisha *(10/2024 – 09/2026)*
Bachelor's Degree in Computer Science — Government Autonomous College, Rourkela, Odisha *(06/2020 – 05/2023)*
Higher Secondary Education (PCMIT) — Vikash H.S., Sambalpur, Odisha *(06/2018 – 03/2020)*
High School — Saraswati Vidyamandir, Sundergarh, Odisha *(06/2017 – 03/2018)*

CERTIFICATIONS

- Java Full Stack Developer (09/2023 – 05/2024)